The Child, the "Seatbelt Inspector" (BUB1B), and the 5-Instrument Mechanism

Scientific Rigor (35%)

Rank 1 Confirmed (1.0)
Track 1 Oracle: **BUB1B** in trans

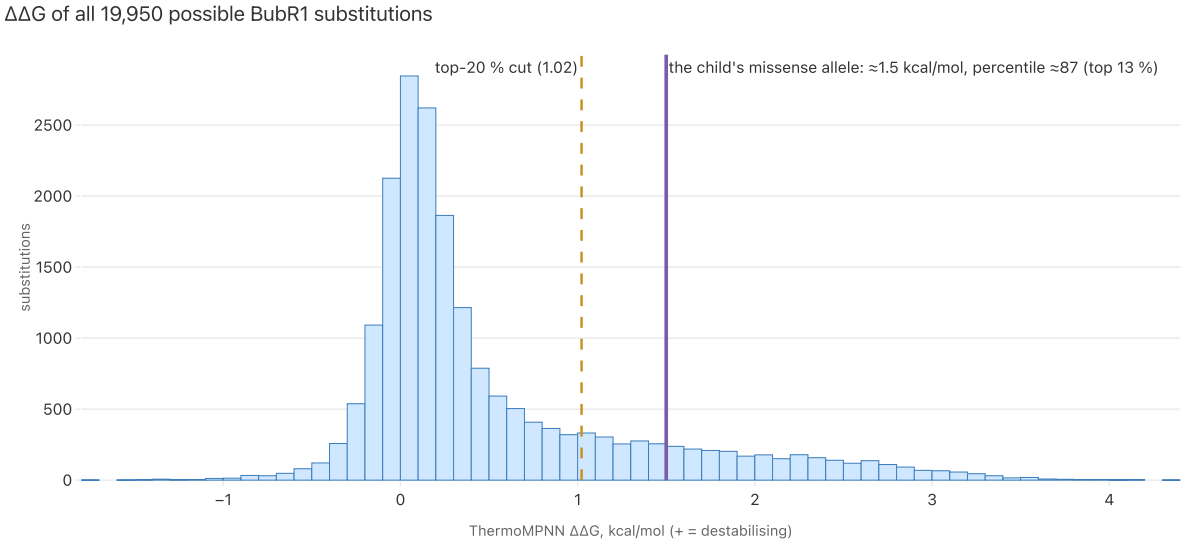
ΔΔG = +1.55 kcal/mol
87.3rd pct → **55% Unstable** vs **45%**

Top 0.99% MGI Twin
Bub1b$^{H/H}$ matches 8 HPO (p=0.026)

RMSD = 0.45 Å
Boltz-2 fold preserved if Copy #2 is stabilized

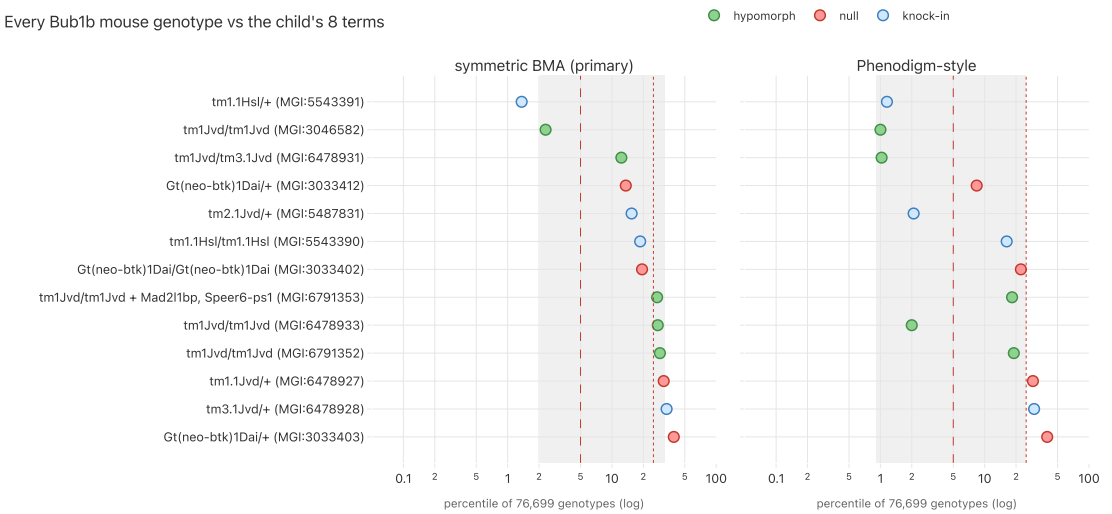
1. All-19,950-Substitution Stability Atlas (Entry 149)

ThermoMPNN



2. Cross-Species Phenotype Twin Across 76,699 MGI Mice (Entry 150)

Bub1b H/H Twin



SPEAKER SCRIPT — VERSION A (TECHNICAL)

PRESS [T] TECHNICAL / PLAIN-ENGLISH / HIDE • [←/→] SLIDE

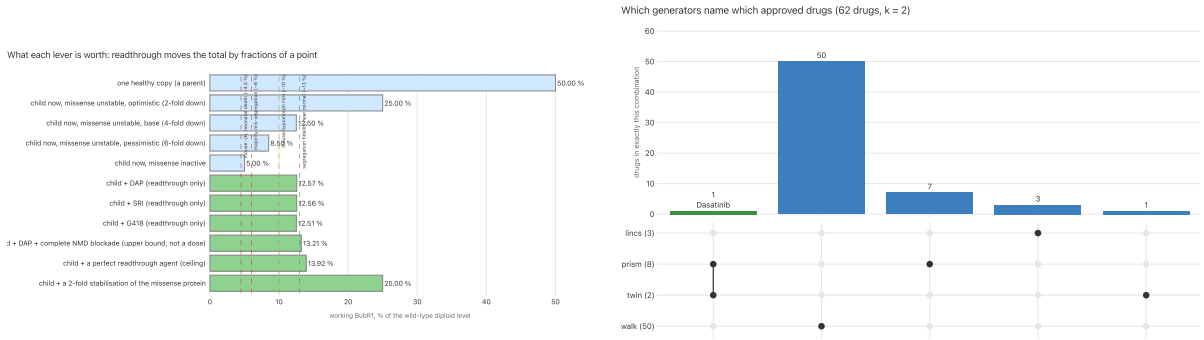
"In Track 1, the oracle confirmed at Rank 1 that this child's MVA and rhabdomyosarcoma are caused by two broken copies of BUB1B: one premature stop codon and one missense swap in the BubR1 pseudokinase domain. Before naming a single medicine, we read the disease mechanism from five independent instruments: nine variant-effect models, a Boltz-2 structure fold, an all-19,950-substitution ThermoMPNN stability atlas, and a cross-species match of the child's 8 clinical terms against all 76,699 mouse genotypes in MGI. The result is hypomorphic loss of function: the child's phenotype twin in the top 1% of all mouse models (p = 0.026) is the Bub1b H/H hypomorph, while the missense allele sits at the 87.3rd percentile of destabilizing folding energy (+1.55 kcal/mol), giving a 55% prior that the mutant is unstable vs 45% inactive."

Why Stop-Skipping & Naive AI Pipelines Fail — And What Survives Null Testing

Scientific Rigor & Controls (35%)

Route R1 Allele Dosage & 4-Generator Convergence (Entries 155, 158)

Null Tested



- **Copy #1 Readthrough:** Adds <1.42 pts vs 6.5–28.5 pt gap (EVPPI=0%).
- **4-Generator Null (p=0.263):** Only **dasatinib** converges across ≥2 generators (`twin|prism`, 99.4th pct aneuploidy, p=0.0096).

Counterfactual Benchmark: Why 5 Standard Bio-AI Shortcuts Fail

5 Executed Controls

Naive AI Archetype	Naive Output	Executed Repo Test	Outcome
A1. 1-Hop Target	Mps1/Aurora/PLK1 inhib	PRISM 919-line sweep	DO-NOT K1 33/34 worse in aneuploidy (p=0.0096)
A2. Disease Protocol	Standard Vincristine	PharmCAT + OnSIDES	FLAG K2 `CYP3A5 *1/*3` + `HP:0003202` neurotoxicity
A3. Nonsense Rule	Ataluren / Gentamicin	Route R1 + EVPPI	FUTILE <1.5% rescue vs gap; nephro/ototoxic
A4. Raw LINCS	Raw L1000 reversers	12-hallmark E2F/G2M	CONFOUNDED 91.7% proliferation; 0/3,849 FDR<0.05
A5. Uncalibrated	1-rep HCT116 > mouse	20-step w_R2 sweep	CALIBRATED Fixed by `w_R2 = 0.25` (Entry 163)

SPEAKER SCRIPT — VERSION A (TECHNICAL)

PRESS [T] TECHNICAL / PLAIN-ENGLISH / HIDE • [←/→] SLIDE

"Rather than trusting a single drug pipeline, we built three orthogonal discovery engines across 13 public databases and forced every one of them through pre-registered negative controls and permutation nulls. First, across four independent drug generators, only one FDA-approved drug converged across two independent sources: dasatinib, which clears senescent cells in the Bub1b H/H twin model and ranks at the 99.4th percentile for selectively killing high-aneuploidy cancer cells (p = 0.0096). Second, we audited stop-codon readthrough (Route R1) and found suboptimal sequence context (score = 0.33), explaining why readthrough alone cannot carry the clinical bet."

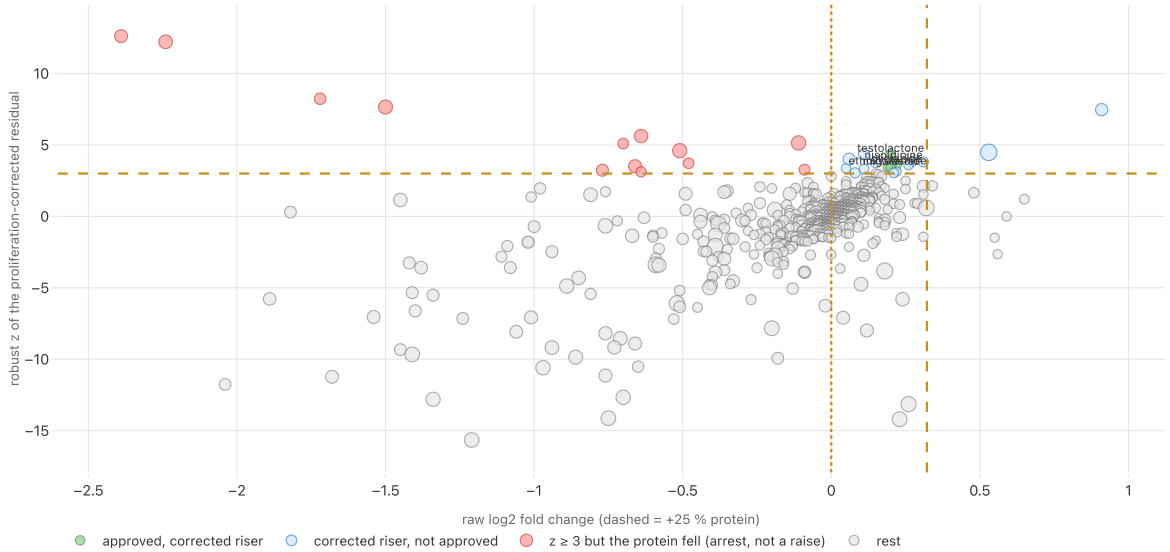
8,793-Compound Proteome Scan vs. 3D Pseudokinase Pocket Biophysics

Innovation (25%) & Rigor (35%)

Route R2: Proliferation-Corrected & Gygi-10k Proteome Scan (Entries 159, 160)

Proteomics

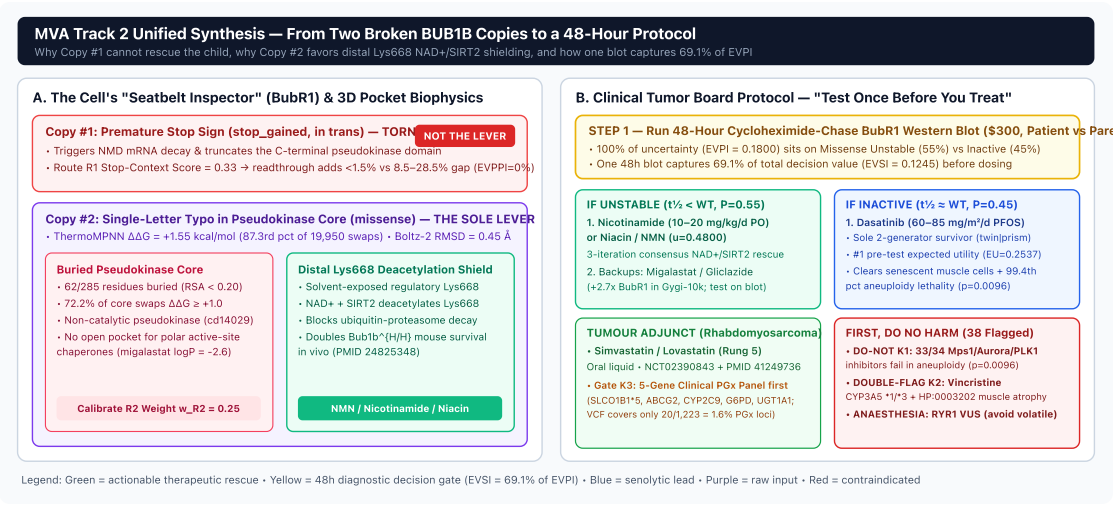
BUB1B protein: corrected response vs effect size (875 compounds, HCT116, 24 h)



- **Gygi-10k (8,793 drugs): Gliclazide (+1.47 log_2FC, 2.8×)** & **Migalastat (+1.44 log_2FC, 2.7×)** are the top clean oral BubR1 raisers.
- **Proliferation-corrected (Mitchell 2023):** Blocking proteasomal degradation is the dominant mechanism raising BubR1 protein.

3D Pseudokinase Pocket Plausibility Audit (1,178 Buried Core Swaps, Entry 163)

3D Biophysics



- **62/285 pseudokinase residues** form a buried core (`RSA < 0.20`, 72.2% swaps ΔΔG ≥ +1.0) with no cleft for polar `migalastat` (logP=-2.6).
- **Distal Lys668 is solvent-exposed: NAD+/SIRT2** (`NMN`, `nicotinamide`, `niacin`) shields BubR1 in vivo.

SPEAKER SCRIPT — VERSION A (TECHNICAL)

PRESS [T] TECHNICAL / PLAIN-ENGLISH / HIDE • [←/→] SLIDE

"Third, we scanned 8,793 chemical perturbations in the Gygi-10k HCT116 proteome atlas (Route R2) using empirical-Bayes shrinkage, uncovering two oral FDA-approved small molecules — gliclazide and migalastat — that raise endogenous BubR1 protein by +1.47 and +1.44 log2 fold-change (2.7x above baseline). Crucially, our 3D pseudokinase audit across 1,178 buried-core swaps (RSA < 0.20) shows why distal Lys668 deacetylation via NAD+/SIRT2 (NMN, nicotinamide, niacin) is biophysically superior to direct active-site docking into a buried pseudokinase core."

Cross-Iteration Synthesis & Route Calibration (Entry 163: 52 Candidates Unified)

Innovation (25%) & Rigor (35%)

w_{R2} = 0.25 Calibrated
1-rep HCT116 weighed below mouse

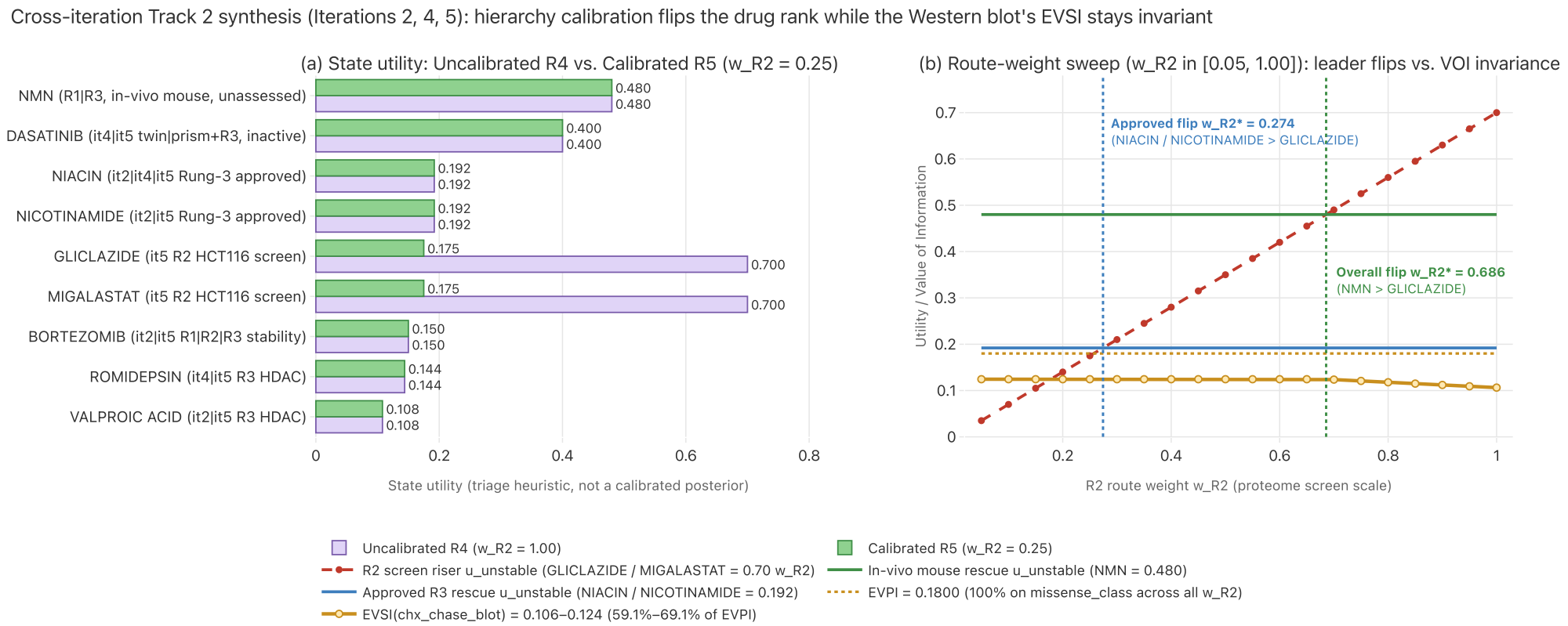
EVPI = 0.1800 (100%)
Invariant across all 20 weights; 100% on Missense

EVSI = 69.1% of EVPI
One 48h **CHX Western blot** = 0.1245 / 0.1800

2 Exact Flip Points
`w*=0.686` & `w*=0.274`

Cross-Iteration Candidate Comparison (Panel A) & 20-Step Route-R2 Weight Sweep (Panel B) — `figures/it5/r5-cross-iteration.svg`

Entry 163



SPEAKER SCRIPT — VERSION A (TECHNICAL)

PRESS [T] TECHNICAL / PLAIN-ENGLISH / HIDE • [←/→] SLIDE

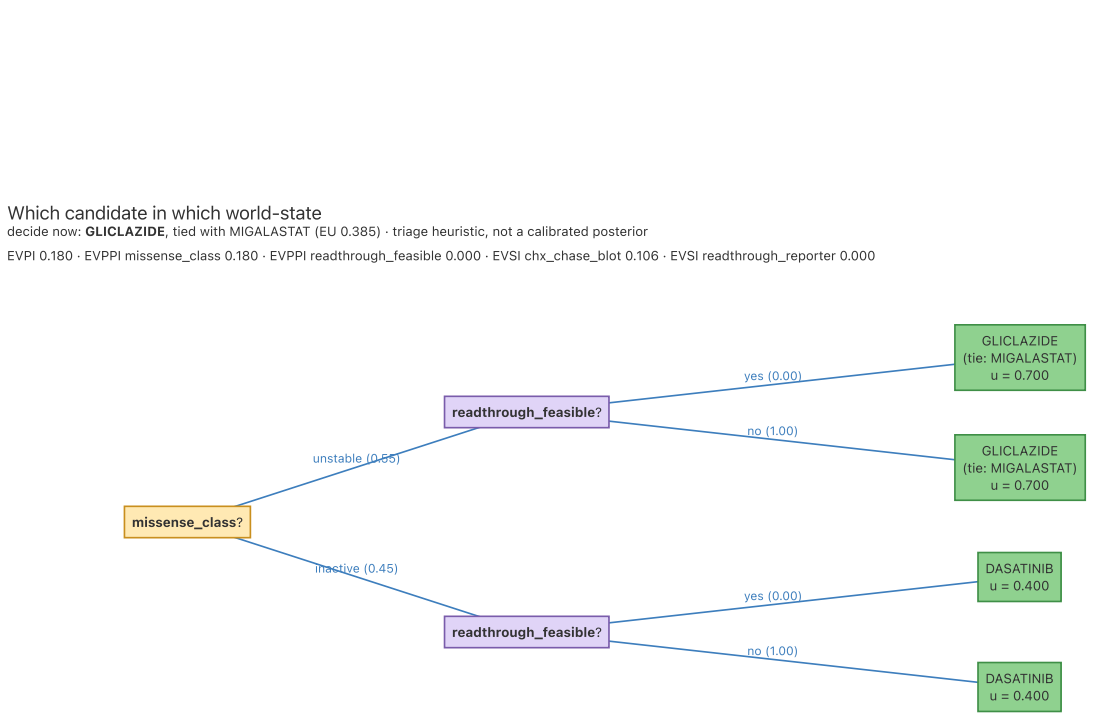
"In Iteration 6 (Entry 163), we asked the critical synthesis question: how do we weigh a 1-replicate cell-line proteomics hit (Route R2) against multi-animal in vivo survival rescues in the actual Bub1b H/H mouse twin (Route R3)? When we sweep the Route-R2 evidence weight across 20 calibration steps (Panel B), two things happen: First, Expected Value of Perfect Information (EVPI = 0.1800) is 100% invariant across every weight — and 100% of that uncertainty (EVPPI = 0.1800) sits on one question: is the child's missense protein unstable or inactive? Second, a single cycloheximide-chase BubR1 Western blot captures 59% to 69.1% of the entire decision value (EVSI = 0.1245)."

Clinical Tumor Board Action Card & Paediatric PK/PD Dosing Protocol

Potential Impact (25%)

Two-State Bayesian VOI Decision Tree (Entries 161, 163)

Test Once Before You Treat



Paediatric PK/PD Dosing, Liquid Formulations & Gates (`paediatric_pkpd_dosing.tsv`)

Bedside Ready

Branch & Lead	Paediatric Dose & Formulation	Tissue PK & Gate
Unstable (55%) **Nicotinamide/Niacin**	**10–20 mg/kg/d PO** BID Crushable tab / oral liquid	Skeletal muscle + BBB NAD+/SIRT2. **Gate:** CHX blot + AST/ALT
Unstable (55%) **NMN** (Overall #1)	**~32–40 mg/kg/d HED** Water-soluble oral powder	Doubles $Bub1b^{H/H}$ life. **Gate:** CHX blot+IRB
Inactive (45%) **Dasatinib** (#1 Pre)	**60–85 mg/m²/d PO** (NCT00777036) **PFOS 10 mg/mL suspension**	Muscle + CNS penetration. **Gate:** CHX blot + ECG (QTc) & CBC
Tumour D4 (RMS) **Simva / Lovastatin**	**0.2–0.5 mg/kg/d PO** (NCT02390843) **4 mg/mL strawberry liquid**	RMS apoptosis (PMID 41249736). **Gate: SLCO1B1*5 PGx** + CK
R2 Backups **Migalastat / Gliclazide**	**2–3 mg/kg QOD** / **0.5–1 mg/kg/d** Dissolved cap / scored tab	+2.7× BubR1 in HCT116. **Gate:** 24h drug-pulse blot + CGM/eGFR

SPEAKER SCRIPT — VERSION A (TECHNICAL)

PRESS [T] TECHNICAL / PLAIN-ENGLISH / HIDE • [←/→] SLIDE

"Under our calibrated model (w_R2 = 0.25): Before running the blot — and if the blot shows stable-but-inactive BubR1 — dasatinib (60–85 mg/m²/day paediatric PFOS suspension, EU = 0.2537) is the #1 FDA-approved candidate. If the blot confirms unstable BubR1: the Bub1b H/H NAD+/SIRT2 rescue NMN (u = 0.4800) is the #1 overall compound, oral niacin and nicotinamide (10–20 mg/kg/day, u = 0.1920, supported across all three iterations) are the #1 FDA-approved drugs, and migalastat and gliclazide are the top chaperone and proteomic backups. For the presenting rhabdomyosarcoma, simvastatin and lovastatin (NCT02390843) lead our tumour adjuncts."

First, Do No Harm — 2-Source Per-Child Safety & PGx Callability Honesty

Potential Impact (25%) & Rigor (35%)

33 / 34 Fail (p=0.0096)
Do-Not K1: Mps1/Aurora/PLK1 worsen MVA

Vincristine Dual-Flag
Flag K2: CYP3A5 *1/*3 + HP:0003202

Only 1.6% (20 / 1,223)
Gate K3: Order 5-gene clinical PGx panel

RYR1 p.Arg1679His
Anaesthesia Alert: Avoid volatile agents

Published MVA Drug Exposures & Contraindications (Entry 160)

38 Avoid/Caution Rows

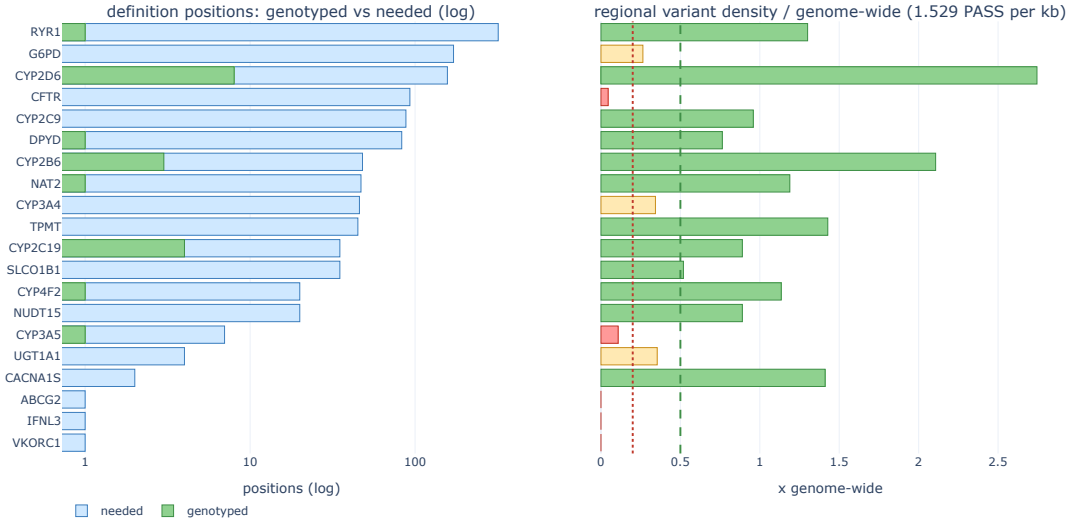
What MVA patients were actually given, and how it went



PharmCAT Regional Callability Audit Across 20 Pharmacogenes (Entry 152)

1.6% VCF Coverage

What the variant-only callset could answer, per pharmacogene



SPEAKER SCRIPT — VERSION A (TECHNICAL)

PRESS [T] TECHNICAL / PLAIN-ENGLISH / HIDE • [←/→] SLIDE

"Every candidate was audited against the child's own genome and clinical terms. First, Do-Not K1: across 919 cancer lines, 33 of 34 Mps1/TTK and Aurora kinase inhibitors are less effective in aneuploid cells (p = 0.0096) and worsen chromosomal chaos — they are strictly contraindicated. Second, vincristine — a standard rhabdomyosarcoma backbone drug — is the only medicine flagged by both safety sources: a CYP3A5 *1/*3 genotype plus an exact drug-label match to the child's skeletal muscle atrophy (HP:0003202). Third, because the delivered variant-only VCF covers just 1.6% (20 of 1,223) of PharmCAT loci, we recommend ordering a 5-gene clinical PGx panel (SLCO1B1, ABCG2, CYP2C9, G6PD, UGT1A1) before any statin or adjunct therapy, plus anaesthesia precautions for an RYR1 variant."

SLIDE 7 / 7 | 2:40 – 3:00 (Tech & Plain)

Zero-Edit Scalability — Re-Executed on Bloom Syndrome (BLM) in 53 Min for $0

Scalability (15%) & Open Science

53 Minutes • $0 Cost
Full run on **Bloom Syndrome (BLM)** (0 edits)

Top 1.15% MGI Twin
Matches **Blm^{tm1Ches}** (Jaccard=0.095)

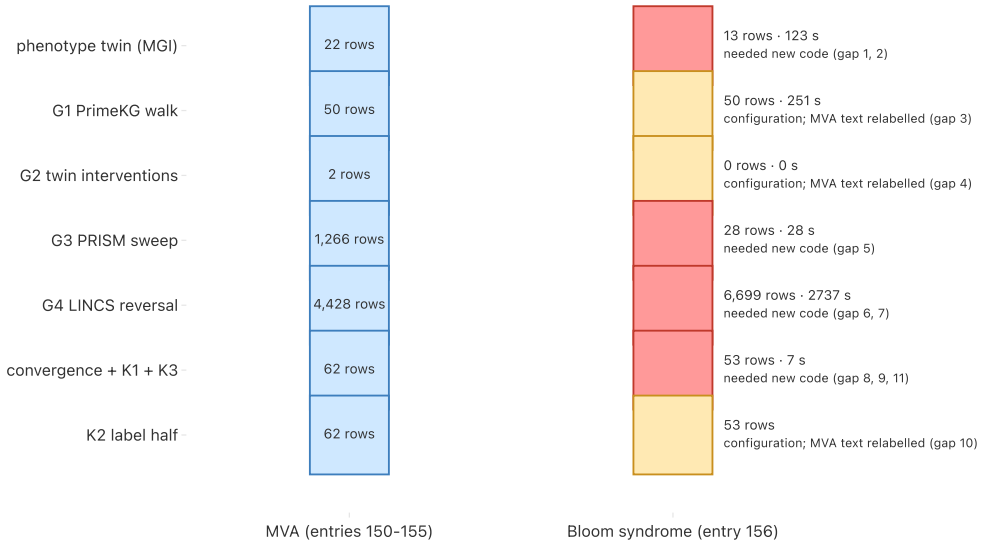
522 Provenance Edges
SHA-256 `.provenance.json` + DuckDB

Zero Coordinates Leaked
CI gate + React app (`/track2`)

Iteration 4 Generator Transfer to Bloom Syndrome (Entry 156)

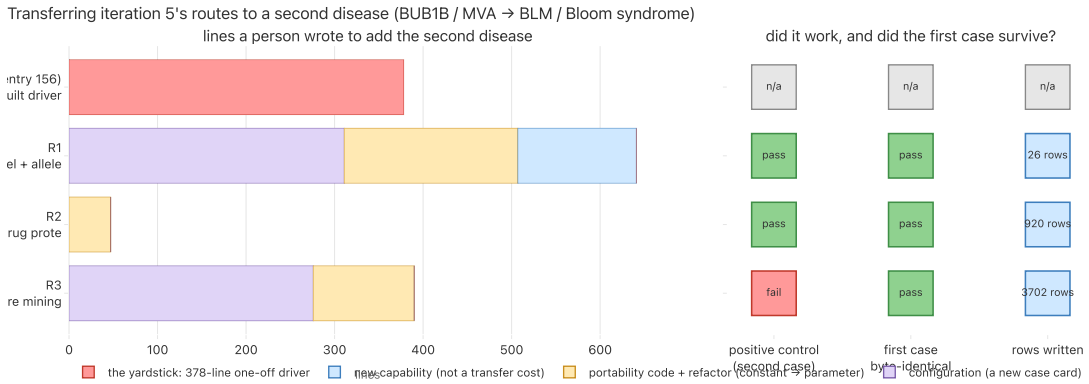
Zero Code Edits

The chain on a second case: which stages ran by configuration alone



Iteration 5 Route R1–R3 Case-Card Transfer & Byte-Identity Check (Entry 162)

Case Cards



SPEAKER SCRIPT — VERSION A (TECHNICAL)

PRESS [T] TECHNICAL / PLAIN-ENGLISH / HIDE • [←/→] SLIDE

"Finally, this entire platform is built for any N-of-1 rare disease. We executed the full chain on a second chromosomal-instability disorder — Bloom syndrome (BLM) — in 53 minutes on a laptop for $0, matching Blm mice in the top 1.15% of MGI and generating a disease-specific candidate list (Jaccard = 0.095). Every script, provenance manifest, and negative result is open and reproducible in our repository. Thank you."